

Course Information

Course Title: Business Analytics in Healthcare Management

Level: Graduate (Master's)

Time: 6:00 PM - 8:15 PM (Wednesdays)

Term: Spring 2026

Instructor: Sangsuk Yoon (syoon1@udayton.edu)

Format: Online Synchronous (Use the Zoom link on Canvas to join the class)

Office Hours: Wednesdays 1:30 PM - 3:00 PM or after class or by appointment (Zoom)

Course Schedule: Please check the course [schedule](#) page

Course Description and Objectives:

Course Description:

This graduate-level course provides a comprehensive introduction to business analytics in healthcare management, progressing from basic statistical concepts and data visualization to practical applications of data analytics in healthcare management contexts. As healthcare organizations increasingly rely on data-driven insights, this course prepares students to navigate and analyze complex healthcare datasets. Students will gain hands-on experience with classic and modern analytical tools while learning to address real-world healthcare management challenges through evidence-based decision-making.

Prerequisites:

There are no prerequisites for this course. This course is designed to accommodate students with varying levels of statistical and programming backgrounds. All necessary concepts will be introduced from the ground up.

Course Objectives:

By the end of this course, students will be able to:

- Evaluate the role and application of analytics in healthcare management decision-making
- Conduct exploratory data analysis on different types of healthcare data (patient surveys, health outcomes, etc.)
- Select and apply appropriate statistical methods to healthcare management problems to extract meaningful insights
- Interpret analytical results and communicate findings effectively to diverse stakeholders
- Design and implement data-driven solutions for real-world healthcare management challenges

Required Materials

Course Website:

1. Course Materials will be posted here and through our Canvas page: <https://mba642-spring-26.github.io>
2. Assignments, grading, and exams will be informed, managed, and submitted through our class Canvas page.

Textbooks:

Two textbooks will be used for this course. You do not need to purchase them. Both books are freely available through the University of Dayton library.

1. [“Statistics for Health Data Science: An Organic Approach”](#)
 - **Authors:** Ruth Etzioni, Micha Mandel, and Roman Gulati
 - **ISBN:** 978-3-030-59888-4
 - This book is freely available through UD library
2. [“Statistical Analytics for Health Data Science with SAS and R”](#)
 - **Authors:** Jeffrey Willson, Ding-Geng Chen, and Karl E. Peace
 - **ISBN:** 978-1-032-32569-9
 - This book is freely available through UD library

Other Helpful Books and Readings:

3. [“R for Data Science”](#) (available for free online)

- **Authors:** Hadley Wickham and Garrett Grolemund
 - This book is freely available online
4. [“Introduction to Modern Statistics”](#)
- **Authors:** Mine Çetinkaya-Rundel and Jeffrey S. Simonoff
 - This book is freely available online
5. [Statistical Rethinking for the 21st Century](#)
- **Authors:** Russell A. Poldrack
 - This book is freely available online

R and R-Studio:

This course utilizes R and R-Studio as the primary analytical tools. R is a powerful, open-source programming language specifically designed for statistical data analysis, widely used in both academia and industry for analytics. R-Studio is an integrated development environment (IDE) that makes R programming more accessible and efficient through its user-friendly interface, offering features such as data visualization, code editing, debugging, and project management capabilities. Students will learn to use these tools to analyze healthcare data, create visualizations, develop statistical models, and generate reports. You can download R and R-Studio from the following links:

- R: <https://www.r-project.org>
- R-Studio: <https://posit.co/downloads/>

You should install R first since R-Studio runs on R.

Grading Components:

Component	Points
Assignments	90 points
Class Participation and Reflection	78 points
Midterm Exam	50 points
Final Exam	50 points
Total	268 points

Assignments: 90 points (9 assignments, each with 10 points)

There will be 9 assignments throughout the semester. Each assignment requires students to apply the analytical methods learned in the class session to new healthcare data analytics scenarios. For example, in Week 4, after we cover data visualization, you will be asked to visualize results with a new dataset. Assignments will focus on different healthcare management contexts and typically take 30-60 minutes to complete. Assignments are due weekly by Sunday at 11:59 PM.

Class Participation and Reflection: 78 points

This course includes 14 class sessions. After we finish each session (starting from the second class, meaning you will submit 13 reflections total), I would like to hear about what you learned and what questions remain unclear. To encourage engagement with the course contents, you will need to write a short list of responses to the following questions:

1. **What were the three most interesting or exciting things you learned from the session? and why?**
2. **What were the three most unclear or confusing things from the session? and why?**

You can include more than three for each, but you must include at least three per question. So, there are six interesting or unclear things you will have in each reflection, and each one will be worth 1 point (total 6 points per class session; max is 6 points). There will be 13 reflection assignments throughout the semester ($6 \text{ points} \times 13 \text{ reflections} = 78 \text{ points}$). Reflection submissions are due weekly by Sunday at 11:59 PM.

You don't need to get stressed about this. Your reflections do not need to be fancy or very long (just one short paragraph for each point). I simply want to understand what genuinely resonated with you or what felt challenging. For example, you might write something like: "Learning how to draw a bar graph using ggplot was interesting. I didn't know how to draw graphs in R before, so discovering this was exciting." Or you could say: "Understanding the interaction effect was confusing. It's unclear to me whether it means the effect of A is different from the effect of B, or whether the effect of A changes depending on B." These kinds of honest, straightforward reflections are exactly what I'm looking for. The goal is to help me understand your learning experience so I can better support you throughout the course.

Midterm Exam: 50 points

There will be a midterm exam in Week 9 of the semester. The exam will cover topics from Week 1 through Week 8. The exam will be online and available for the full day on the scheduled class day (from 12:00 AM to 11:59 PM).

Final Exam: 50 points

There will be a final exam in the last week of the semester. The exam will be cumulative and cover topics discussed throughout the semester. The exam will be online and available for the full day on the scheduled class day (from 12:00 AM to 11:59 PM).

Final course letter grades will be assigned as follows:

Letter Grade	Quality Points	Point Range
A	4.000	250 - 268
A-	3.667	240 - 249
B+	3.333	230 - 239
B	3.000	220 - 229
B-	2.667	210 - 219
C	2.000	185 - 209
F	0.000	0 - 184

Class Policies

Attendance Policy:

This course is delivered online synchronously, and regular attendance is expected. Attendance will be recorded at a random point during each session. I understand that life happens. If you must miss a class, please let me know in advance when possible. All lecture recordings and materials will be available on Canvas so you can catch up. However, missing class does not relieve your responsibility for assignments, reflections, or exams, as these are due on set deadlines regardless of attendance.

Communication Policy:

- Feel free to reach out to me anytime at my UD email: syoon1@udayton.edu. I'm always happy to help!
- I typically respond to emails within 24 hours on weekdays (Monday–Friday).
- On weekends and holidays, I'll be recharging my batteries (and pretending emails don't exist), so expect a response when I'm back online. Thanks for understanding! :)

Classroom Etiquette:

- **Camera on, please!** Keeping your camera on helps create a more engaging and connected learning environment.
- Think of our class sessions as professional team meetings, which is a valuable practice for your current and future career.
- Be respectful to yourself, your peers, and our community. Let's maintain a welcoming and professional atmosphere.
- Use professional language in all class discussions and written work.

Technological Problems:

- Please ensure you have reliable internet access for joining class and submitting assignments.
- Keep your UD email, Zoom, and Canvas accounts active and accessible throughout the semester.
- You are responsible for resolving any technical issues with your personal devices or software (including R and RStudio).
- **Need tech help?** For hardware issues, contact your device manufacturer. For general software issues, reach out to UD IT services. For course-specific technical questions (like R code not working), feel free to ask me!

Late Submissions:

Life gets busy, and I understand that sometimes things don't go as planned. However, please try your best to submit on time. Here's what happens if you're late:

- **Reflections:** 1.5 points deducted for each 24 hours late. Submissions over 48 hours late receive zero points.
- **Assignments:** 2 points deducted for each 24 hours late. Submissions over 48 hours late receive zero points.
- **Exams:** Must be completed within the specified time frame—no late submissions accepted.

Keep Your Work:

- Always save a copy of your submitted assignments—just in case anything gets lost along the way.
- Hold onto your graded work as well. If you ever have questions about a grade (e.g., a mismatch between Canvas and your feedback), having the original document will help us resolve it quickly.

Stay on Top of Your Grades:

Please check your grades on Canvas regularly. If you notice something missing or think a grade might be incorrect, don't hesitate to reach out. I'm happy to take another look!

University Policies

Civility, Respect, and Privacy

The University of Dayton adopted a "Statement of Dignity which states the following: "A primary assertion of both our religious and civil traditions is the inviolable dignity of each person. Recognition of and respect for the person are central to our life as a Christian and education community and are what allow us to pursue our common mission while being many diverse persons. Thus discrimination, harassment, and any other conduct that diminishes the worth of a person are incompatible with our fundamental commitment as a Catholic university conducted in the Marianist tradition."

Classroom Climate and Bias Experiences and Process

As part of a University for the Common Good that aims to be an anti-racist university, I want all students to feel safe, respected, and valued in all of their classes. Should you have an experience in this class, or in any SBA class, where you feel devalued because of an incident in class, we would like to know. You are free to discuss the issue with me. You can also contact our Associate Dean, Dr. Marsha Keune (mkeune1@udayton.edu), share your experience with the Equity Compliance Office (equitycompliance@udayton.edu or 937-229-3622) or submit an incident report.

Intellectual Property Statement

The materials shared with you during this course are authored and owned by the instructor and/or the book publisher. Copyright laws must be respected at all times.

Academic Honesty

You are expected to do your own written work on all assignments and projects. For specific UD policies concerning academic honesty, see the University's Academic Honor Code in the Academic Catalog.

Contract Cheating

Contract cheating occurs when a student pays someone (such as a person or company found online) to write essays or complete assignments for a class. This defeats the purpose of the course, and I encourage you to follow the academic honor code and seek to develop a personal sense of academic integrity which lays the foundation for an ethical life. Contract cheating is especially dangerous because of the risk of bribery and extortion (at any point after college.) Students are often lured into contract cheating by signing up for online tutoring services. Remember that UD provides many learning support resources through the Office of Learning Resources.

Academic Integrity and ChatGPT/AI Generated Content

As a student at UD, you are expected to adhere to the Standards of Conduct outlined in the Academic Honor Code. One of the goals for your educational development is to develop your critical thinking and writing skills, which can be weakened by regularly utilizing AI for writing. Thus, for this course, all submitted writing should be your own, original work. If you quote or copy content from ChatGPT or other AI sources, it must be appropriately cited in your submissions. As your instructor, I am required to investigate all suspected violations of academic dishonesty and follow the honor code violation procedures, which can result in an NC for the course.

Dropping the Course

You are responsible for understanding the University's policies and procedures regarding withdrawing from courses. You should also be aware of the current deadlines and penalties for dropping classes. Information on withdrawal from courses is available in the Academic Catalog under Grades and Scholarship. You may also want to speak with your academic advisor.

Student Feedback Tool

The University will ask for your anonymous feedback regarding instruction in this course through the online Student Feedback Tool (SFT). Your candid, respectful opinions and constructive suggestions have an impact on the quality of teaching at UD. Instructions for how to complete SFT will be sent to your UD email account towards the end of the term, and you will be given time in class to complete it. If you encounter teaching problems accessing SFT, contact the UDiT Service Center at 937-229-3888 or itservicecenter@udayton.edu. To learn more about SFT, visit <https://go.udayton.edu/sft>.

University Services and Resources

It's the University's goal that learning experiences be as accessible as possible. Your learning in class is important to me, so I want you to be aware of campus resources which are available to all UD students.

Support for Your Learning in This Course

The Ryan C. Harris Learning Teaching Center's Office of Learning Resources (OLR) is a learning resource for students, parents, faculty, and staff at the University of Dayton. Please contact OLR via email at OLR@udayton.edu or by calling 937-229-2066 (TTY: Ohio Relay 711 for deaf/hard of hearing individuals) or visit the office on the ground floor of Roesch Library (LTC 023) if you would like to talk about how you could become a more effective learner. You can also check out the website: go.udayton.edu/learning.

Students with Disabilities

If you are a student with a disability and feel you may need a reasonable accommodation to fulfill the essential functions of this course, you are encouraged to connect with the Office of Learning Resources (OLR). OLR provides accommodations and services for students with a variety of disabilities, including physical, medical, and psychiatric. You may contact OLR to learn more about the process and procedures for requesting accommodations via email at disabilityservices@udayton.edu or at 937-229-2066 (TTY: Ohio Relay 711 for deaf/hard of hearing), or by visiting the office in Roesch Library (LTC 023.)

If you need assistance with alternate formats of electronic course materials (such as converting a PDF into an accessible format), visit OLR's website to use the tool, Sensus Access, for document conversion.

Wellbeing and Success

At UD, we know health and wellbeing are important to your success. That's why we help students of all backgrounds build a foundation for holistic wellbeing. We provide the services, resources and support you need to be successful on campus and in life. Health and Wellbeing resources include the following:

- Dean of Students Office: Student Care and Advocacy at 937-229-1212
- Counseling Center at 937-229-3141
- Student Health Center at 937-229-3131
- Center for Alcohol and Other Drugs Resources and Education at go.udayton.edu/cadre
- Brook Center at go.udayton.edu/brook (includes LGBTQ+ Student Services)

- Campus Recreation at activeflyers.udayton.edu
- YOU@Dayton at wellbeing.udayton.edu

Student Success Network

UD makes student success a priority. For this reason, I will be using the Student Success Network (SSN) in this course as an early identification and intervention system designed to enable your academic success and graduation. If I notice that you are struggling with issues such as attendance, class participation, or assignment performance, I may choose to send notification through SSN in order to connect you with appropriate campus resources. These referrals are designed to maximize your chances for success, not as punishment, so please respond to any communications you may receive from me, your Academic Advisor, the Dean's Office, or other campus offices regarding your academic progress in this course. Please see the Student Success Network website for more information.

Write Place

The Write Place offers free writing assistance on any assignment at any stage of the writing process. To make an appointment with the Write Place, email writeplace@udayton.edu.

Library Research Support

Roesch Library's reference services offer free research assistance on any assignment at any stage of the research process. Visit the first floor Information Desk to meet in-person, or connect remotely with a librarian via the Libraries' Get Help page. Research consultation service hours are listed online. No appointment is necessary, although they are available. To make an appointment with a librarian, see either the Library Resources tab in Canvas or make an appointment online.

Religious Accommodations

UD strives for an inclusive climate and welcomes students from all backgrounds, faiths and experiences. If religious observance impedes your ability to participate fully in classroom activities or a principal holiday from your religious tradition occurs during the semester and conflicts with class, please make me aware of this immediately to determine if a reasonable accommodation is possible. Find out more about how to request a religious accommodation.